

Vansh Virani

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SUMMARY

Software Engineer and incoming Data Science graduate student at SJSU with expertise in architecting **scalable backend services** and **cloud-native ML pipelines**. Proficient in building **RAG systems**, **distributed model monitoring**, and full-stack applications using Python, Java, and React. Demonstrated success in deploying production-grade solutions within **Agile/Scrum** environments and optimizing data-driven workflows for high-user loads.

EDUCATION

M.S., Software Engineering (Specialization: Data Science) – San Jose State University **Aug 2026 – Aug 2028**
• **Planned Coursework:** Data Mining (CMPE 255), Machine Learning (CMPE 257), Deep Learning (CMPE 258), Software Systems Engineering (CMPE 202).

B.S., Computer Science – San Francisco State University **Aug 2021 – Aug 2025**
• **Cumulative GPA:** 3.11
• **Academic Honors:** Dean's List (Fall 2023, Spring 2024, Fall 2024, Winter 2025)
• **Relevant Coursework:** Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Calculus I-III, Deep Learning, Software Engineering, Operating Systems.

TECHNICAL SKILLS

- **Languages:** Python, Java, Kotlin, TypeScript, JavaScript (ES6), SQL, HTML5, CSS3, Bash/Shell Scripting
- **Frameworks:** Django, FastAPI, Flask, Spring Boot, React, Node.js, Express.js, LangChain, PyTorch
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Kubernetes, Jenkins, GitHub Actions (CI/CD), Cloud-native Architecture
- **Databases:** MySQL, PostgreSQL, MongoDB, Redis, FAISS (vector storage)
- **AI/ML:** Generative AI, LLMs, RAG Pipelines, Prompt Engineering, Transfer Learning, CNNs, Model Deployment
- **Monitoring & Observability:** Splunk, Grafana, Prometheus, Datadog, Google Analytics, Logging Pipelines
- **Tools:** Agile/Scrum, REST APIs, Microservices Architecture, Test-Driven Development, Git, JUnit, Postman, QA Automation

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | [Link](#)
- Oracle Cloud Infrastructure (OCI) 2025 - AI Foundations Associate | [Link](#)

PROJECTS

GoldenPath – Full-Stack Tutoring Platform | *Python, Django, MySQL, AWS, Ubuntu*

- **Collaborated** in a **cross-functional Agile team** of 7 to **design, build, and deploy** a tutoring platform serving **30,000+ users** — contributing across the **full software development lifecycle**.
- **Designed** a **scalable relational database schema** using **Django ORM** with denormalization strategies, enabling real-time rating computations with **sub-second API response times**.
- **Deployed** the **production environment** on **AWS EC2** with Apache, configuring **high-availability** infrastructure to support **concurrent user sessions** reliably.
- **Implemented** secure **RESTful APIs** enforcing campus-domain **email authentication** and **password hashing**, following enterprise **security best practices**.
- Integrated **Splunk-based logging** and **Google Analytics** for **system observability**, enabling proactive **debugging** and **data-driven** performance improvements.

Savaalish – Generative AI Simulation Pipeline | *Python, FastAPI, LangChain, FAISS*

- Architected a **cloud-native data pipeline** to evaluate **generative AI model behavior** through automated knowledge-task simulations.
- Engineered **token-aware chunking** and **FAISS vector retrieval** to implement a production-ready **Retrieval-Augmented Generation (RAG)** pipeline.
- Built **RESTful APIs** for reproducible experimentation, following **microservices design patterns** for modular pipeline orchestration.

Skin Cancer Classification & ML Ops | *Python, PyTorch, Docker, Kubernetes*

- Developed a **cloud-native ML pipeline** to classify dermatoscopic images using **CNN architectures (ResNet, EfficientNet)**.
- **Containerized** with **Docker** and **orchestrated** on **Kubernetes** for **scalable distributed inference**.
- Integrated **Prometheus** and **Grafana** dashboards to monitor **model drift** and **latency**, improving experiment traceability by 40%.

Gator Goods – Campus Marketplace SPA | *React, Node/Express, MySQL*

- **Designed and developed** an **end-to-end** e-commerce **single-page application** with **RESTful APIs** enabling listing management, order processing, and real-time status tracking.
- **Optimized SQL query performance by 35%** through **indexed schemas** and debounced search filters, ensuring **scalable** data retrieval under **concurrent load**.
- Led **version control** and **code review** practices on **GitHub**, enforcing branching strategy and conducting **defensive testing** to **validate data integrity** across **edge cases**.